

Chapter No.1

The Scope of Microbiology

QUESTIONS:

Q. 01. List the characteristics common to all biological systems?

ANS: The 7 Characteristics of Life:

1. Living Things Are Composed of Cells:

- Single-cell organisms have everything they need to be self-sufficient.
- In multicellular organisms, specialization increases until some cells do only **certain things**.

2. Level of organization:

Living things have different level of organization which includes both molecular and cellular organization.

- Living things must be able to organize simple substances into complex ones.
- Living things organize cells at several levels:
- **Tissue** - a group of cells that perform a common function.
- **Organ** - a group of tissues that perform a common function.
- **Organ system** - a group of organisms that perform a common function.
- **Organism** - any complete living thing.

3. Living Things Use Energy:

Living things take in energy and use it for maintenance and growth.

4. Living Things Respond to Their Environment:

- Living things will make changes in response to a stimulus in their environment.
- A behavior is a complex set of responses.

5. Living Things Grow:

- Cell division - the orderly formation of new cells.
- Cell enlargement - the increase in size of a cell. Cells grow to a certain size and then divide.
- An organism gets larger as the number of its cells increases.

6. Living Things Reproduce:

- Reproduction is not essential for the survival of individual organisms, but must occur for a species to survive.
- All living things reproduce in one of the following ways:
 - Asexual reproduction - Producing offspring without the use of gametes.
 - Sexual reproduction - Producing offspring by the joining of sex cells.

7. Living Things Adapt to Their Environment:

- Adaptations are traits giving an organism an advantage in a certain environment.
- Variation of individuals is important for a healthy species.

Q. 02. Why is microorganism useful as subject for research in the field of biology?

ANS:

- Microbes are useful tools in research because of their rapid life cycle, their simple growth requirements and their small size.
- Due to this simplicity, microbes have essential in understanding core questions in biology.
- Attempts to classify microorganisms have led to a classification system that divides all organisms into three domains: Archaea, Bacteria and Eukaryote.
- Microbes provide tools for use in molecular biology. These tools have allowed scientists to make rapid progress in investigating many types of microorganisms.

Microorganisms used in research have many useful properties. They grow on simple, cheap medium and often give rise to large populations in a matter of 24 hours. It is easy to isolate their genomic material, manipulate it in the test tube and then place it back into the microbe. Due to their large populations it is possible to identify rare events and then, with the use of powerful selective

techniques, isolate interesting bacterial cells and study them. These advantages have made it possible to test hypotheses rapidly. Using microbe's scientists have expanded our knowledge about life.

Q. 03. Explain why a knowledge of microbiology is useful in understanding life processes in higher plants and animals?

ANS: Microorganisms can be found in almost every natural element on the planet. The lives of human, plants and animals are all intrinsically linked to the microbes that continually recycle key nutrients such as carbon and nitrogen, degrade organic matter and shape our day to day existence.

Q. 04. How did the term protists arise? What organisms do we refer to by use of this term?

What is the differences b/w lower protists and higher protists?

ANS: One of the earliest proposals was made in 1866 by a German Zoologist E.H. Haeckel. He suggested that a third kingdom, PROTISTA, be formed to include those unicellular microorganisms that are typically neither plants nor animals. These organisms "the protists" include bacteria, algae, fungi and protozoa. Viruses are not cellular organisms and therefore are not classified as protists. Bacteria are referred to as lower protists; the others algae, fungi and protozoa are called higher protists.

Q. 05. Discuss the differences b/w prokaryotic & eukaryotic cells?

ANS: Difference between prokaryotic and eukaryotic cells is summarized in the table below.

Features	Prokaryotes	Eukaryotes
Origin	Around 3.5 billion years ago	Around 2 billion years ago
Size	0.1–5.0µm in diameter	10-100µm in diameter
Features	Prokaryotes	Eukaryotes
Cellular Organization	Most are unicellular	Most are multicellular

Genetic Material	The genetic material (DNA) is suspended in nucleus the nucleoid.	The genetic material (DNA) is found in the nucleus.
Shape of Genetic Material	DNA is circular	DNA is linear and is packed with histone proteins.
Number of Chromosomes	Only one	Multiple chromosomes are present
Nucleus	No nucleus	Has a membrane bound nucleus
Plasma Membrane	Made up of peptidoglycan	Made up of phospholipids
Cytoplasm	Has cytoplasm	Has cytoplasm
Organelles	No organelles wrapped in membranes	Has organelles wrapped in membranes (e.g. mitochondria, Golgi bodies, smooth and rough ER, vacuoles, etc.)
Cytoskeleton	Present	None
Glycocalyx	Present	Only in some
Ribosomes	Made up of rRNA and proteins	Made up of rRNA and proteins
Flagella	Present	Present
Asexual	Possible through binary fission	Possible through mitosis

Q. 06. How do viruses differ from other microorganisms?

ANS:

Features	Bacteria	Virus
Ribosomes	Present	Absent
Cell wall	Peptidoglycan/ Lipopolysaccharide	No cell wall. Protein coat present instead.
Introduction	Bacteria constitute a large domain of prokaryotic microorganisms. Typically, a few micrometers in length, bacteria have a number of shapes, ranging from spheres to rods and spirals	A virus is a small infectious agent that replicates only inside the living cells of other organisms.
Living attributes	Living organism	Opinions differ on whether viruses are a form of life or organic structures that interact with living organisms.
Nucleus	No	No
Number of cells	Unicellular; one cell	No cells; not living

Q.11. Name several applied areas of microbiology. Describe the importance of microorganism in each of these applied fields?

ANS: Major Fields of applied microbiology are given below in the table:

Field	Some Applied Areas
Medical microbiology	Causative agents of disease, diagnostic procedures; diagnostic procedures for identification of causative agents; preventing measures.
Aquatic microbiology	Water purification; microbial examination; biological degradation of waste; ecology
Aero microbiology	Contamination and spoilage; dissemination of disease
Food microbiology	Food preservation and preparation; foodborne diseases and their prevention
Agriculture microbiology	Soil fertility; plant and animal diseases
Industrial microbiology	Production of medicinal products such as antibiotics and vaccines; industrial chemicals; production of proteins and hormones by genetic engineered microorganisms.
Exo-microbiology	Exploration for life in outer space
Geochemical microbiology	Coal, minerals and gas formation; recovery of minerals from low grade ores.

Chapter No.2

The History of Microbiology

QUESTIONS:

Q.01. What is the relationship between the germ theory of fermentation and the germ theory of disease?

ANS: The fermentation theory of disease is the (now obsolete) concept that many diseases, including the diseases which were "epidemic, endemic and contagious", owe their origin to the presence of a "morbific principle" in the system, acting in a manner analogous to, although not identical with, the process of fermentation. It was rendered obsolete by the germ theory of disease, which led to the new science of bacteriology.

The fermentation theory was studied in depth and brought to light first by Louis Pasteur. This theory states that it is the idea or concept of how fermentation is brought on by microbes and put to the concept of spontaneous generation to rest. Even though this theory is now outdated and has been replaced by the germ theory of disease, for a long time it held true, and Louis was on the forefront of explaining why it seemed organisms appeared out of nothing instead of claiming it was just a spontaneous act of God. Fermentation was a process that has been used for thousands of years, but no one could explain exactly what was happening and why. From Pasteur's discovery of why and how fermentation occurs, the process has been studied intensely and is now a mastered art used in everyday life with processes of making things such as alcoholic beverages, some foods like yogurt or even manufacturing some medications. The germ theory of disease states that many diseases are caused by microorganisms. These small organisms, too small to see without magnification, invade human, animals, and other living hosts. Their growth and reproduction within their hosts can cause a disease. The germ theory was proposed in 1546 by Girolamo Fracastoro and expanded upon by **Marcus von Plenciz** in 1762.

Q. 02. What is meant by pure culture concept? Does it exist in natural environments? Explain.

ANS: The growth of a mass of cells of the same species in a laboratory vessel (such as test tube) is called a **pure culture**. Pure culture of bacteria was first obtained by Joseph Lister in 1878 using serial dilutions in liquid media. The importance of pure culture to the development of science cannot be overestimated, since by using pure culture techniques the microorganisms responsible for many infections, certain fermentations, nitrogen fixation in soil and other activities were isolated and identified.

Q.03. How did Leeuwenhoek's work influence the contribution of Pasteur and Koch? ANS: Pasteur, a French scientist who made great contributions to our understanding of microbiology and for whom the process of "pasteurization" is named, brought to light strong evidence that microbes arise from other microbes, not spontaneously.

Koch, a German physician, developed a sequence of experimental steps for directly relating a specific microbe to a specific disease and in doing so was able to isolate:

Bacillus anthracis (the bacterium causing anthrax)

Mycobacterium tuberculosis (the bacterium causing tuberculosis) *Vibrio cholera* (the bacterium causing cholera).

Q. 04. In what way are the contribution of Ehrlich and Fleming related?

ANS: Paul Ehrlich (1854-1915) developed modern concept of chemotherapy and chemotherapeutic agents. Alexander Fleming was a Scottish scientist and microbiologist. At an early age, he began to develop his love for science as a member of the Royal Polytechnic Institute in London. This 'accident' led to the discovery of the first antibiotic, which he called penicillin.

Q. 05. In what did Koch's postulates influence the development of microbiology?

ANS: Koch's postulates are four criteria designed to establish a causative relationship between a microbe and a disease. The postulates were formulated by Robert Koch and Friedrich Loeffler in 1884, based on earlier concepts described by Jakob Henle, and refined and published by Koch in 1890. Koch applied the postulates to describe the etiology of cholera and tuberculosis, but they have been controversially generalized to other diseases. These postulates were generated prior to understanding of modern concepts in microbial pathogenesis that cannot be examined using Koch's postulates, including viruses (which are obligate cellular parasites) or asymptomatic carriers. They

have largely been supplanted by other criteria such as the Bradford Hill criteria for infectious disease causality in modern public health.

Koch's postulates are the following:

1. The microorganism must be found in abundance in all organisms suffering from the disease, but should not be found in healthy organisms.
2. The microorganism must be isolated from a diseased organism and grown in pure culture.
3. The cultured microorganism should cause disease when introduced into a healthy organism.
4. The microorganism must be re-isolated from the inoculated, diseased experimental host and identified as being identical to the original specific causative agent.

Q. 06. Why is the period 1880-1900 significant for the emergence of microbiology as a science?

ANS: The period from 1880-1900 was indeed a golden time for microbiology because the infant science grew to adolescence during those years. Most of the causative agents of bacterial disease that had plagued the world for centuries were isolated and identified.

Q. 07. Why was the introduction of the use of agar important to microbiology? ANS: Agar's chief use is as a culture medium for various microorganisms, particularly for bacteria, its other less well-known uses include serving as a thickening for soups and sauces, in jellies and ice cream, in cosmetics, for clarifying beverages, and for sizing fabrics. Agar is a phycocolloid extracted from a group of red-purple marine algae (Class Rhodophyceae) including *Gelidium*, *Pterocladia* and *Gracilaria*. *Gelidium* is the preferred source for agars. Impurities, debris, minerals and pigment are reduced to specified levels during manufacture. Agar is a gel at room temperature, remaining firm at temperature as high as 65°C. Agar melts at approximately 85°C, a different temperature from that at which it solidifies, 32-40°C. This property is known as hysteresis. Agar is generally resistant to shear forces; however, different agars may have different gel strengths or degrees of stiffness.

Agar is typically used in a final concentration of 1-2% for solidifying culture media smaller quantities (0.05-0.5%) are used in media for motility studies (0.5% w/v) and for growth of anaerobes (0.1%) and microaerophiles.

Specifications for bacteriological grade agar include good clarity, controlled gelation temperature, controlled melting temperature, good diffusion characteristics, absence of toxic bacterial inhibitors and relative absence of metabolically useful minerals and compounds.

Q. 08. For what contribution to microbiology made by each of the following :(a) Erwin F. Smith, (b) Emil Christian Hansen (c) T.J. Burrill?

ANS:

(a) Smith early American microbiologists discovered transmission of Texas fever by cattle tick. (b) Emil Christian Hansen developed important procedure for differential staining of bacteria, the gram stain.

(c) T.J Burrill discovered bacterial diseases of plants.

Q. 09. Name an important contribution to microbiology made by each of the following: (a) Jenner, (b) Metchnikoff, (c) Lister, and (d) Welch.

ANS:

(a) Edward Jenner discovered vaccination for smallpox using cowpox vaccine.

(b) Elie Metchnikoff discovered phagocytosis.

(c) Lister developed aseptic techniques isolated bacteria in pure culture.

(d) Welch, one of first great American microbiologists discovered organisms that cause gas gangrene.

Q. 10. Describe the process of phagocytosis?

ANS: Phagocytosis, process by which certain living cells called phagocytes ingest or engulf other cells or particles. The phagocyte may be a free-living one-celled organism, such as an amoeba, or one of the body cells, such as a white blood cell.

Types of Phagocytes. The particles commonly phagocytized by white blood cells include bacteria, dead tissue cells, protozoa, various dust particles, pigments, and other minute foreign bodies. In humans and in vertebrates generally, the most-effective phagocytic cells are two kinds of white blood cells: the macrophages (large phagocytic cells) and the neutrophils (a type of granulocyte). The macrophages occur especially in the lungs, liver, spleen, and lymph nodes, where their function is to

free the airways, blood, and lymph of bacteria and other particles. Macrophages also are found in all tissues as wandering amoeboid cells, and the monocyte, a precursor of the macrophage, is found in the blood. The smaller phagocytes are chiefly neutrophils that are carried along by the circulating blood until they reach an area of infected tissue, where they pass through the blood vessel wall and lodge in that tissue. Both macrophages and neutrophils are drawn toward an area of infection or inflammation by means of substances given off by the bacteria and the infected tissue or by a chemical interaction between the bacteria and the complement system of blood serum proteins. Neutrophils may also engulf particles after colliding with them accidentally.

Q. 11. How did Ehrlich's concept of diseases differ from the phenomenon of phagocytosis?

ANS: Ehrlich's concept of diseases differs from the phenomenon of phagocytosis because it was based on immunity. Paul Ehrlich explained immunity on the basis of certain soluble substances in the blood. We know that there is much merit in both arguments and that both mechanisms play their parts. Ehrlich made important discovery that opened the door to future development in chemotherapy and antibiotics: he found that the 606th compound tested, an organic chemical compound containing arsenic, would destroy the syphilis microbe in the body. This was the first chemotherapeutic substance scientifically discovered and evaluated.

Q. 12. Many researchers in the biological sciences use microorganisms as a model system to explore life processes. Explain why this is so.

ANS: As new laboratory techniques and experimental procedures were developed, knowledge of the characteristics of microorganisms accumulated rapidly. Extensive information about the biochemical activities of microorganisms became available. An analysis of the data suggests that there was much in common among different microorganism. At about the same time there was a growing recognition of the unity of the biochemical processes in microorganisms and higher forms of life, including human being. Consequently, it became attractive to use microorganisms as a tool to explore fundamental life processes. Microorganisms offer numerous advantages: they reproduce very rapidly, they can be cultured in small or vast quantities conveniently and rapidly, their growth can be manipulated easily by chemical or physical means and their cells can be broken apart and the contents separated into fractions of various particles sizes. These as well as other characteristics, make organisms very convenient research models.

Q. 13. For what contributions are the following microbiologists remembered:

(a) Winogradsky, (b) Hellriegel and Wilfarth, (c) Beijerinck?

ANS: (a) In the late 1800s the Russian Sergei Winogradsky, showed the importance of bacteria in taking nitrogen from the atmosphere, combining it with other elements, and making it available as plant food and hence as animal food. (b) In 1888, the mutually beneficial, or symbiotic, relationship between bacteria and the leguminous plant, such as clover and alfalfa was shown by H. Hellriegel and H. Wilfarth. (c) Martinus Willem Beijerinck found the free – living nitrogen – fixing bacterium *Azotobacter* and describe its usefulness in promoting soil fertility.

Q. 14. List some of the applied fields of microbiology and make a general statement of the importance of microorganisms in each field. ANS: The fields of applied microbiology are following:

- Medical Microbiology
- Pharmaceutical Microbiology
- Industrial Microbiology
- Microbial Biotechnology
- Food Microbiology and Dairy Microbiology
- Agricultural microbiology

The microbes are responsible for numerous beneficial processes.

Q. 15. What in your judgment, were the five most important discoveries between 1800 and 1900 which contributes to the establishment of microbiology as a science?

ANS: The five major discoveries are following:

- Development of modern concept of chemotherapy and chemotherapeutic agents.
- Introduction of use of antiseptics.
- Establishment of germ theory of fermentation and germ theory of disease, develop immunization techniques.
- Develop aseptic techniques; isolated bacteria in pure culture.

Chapter No.03

The Characterization, Classification, and Identification of Microorganisms

QUESTIONS:

Q. 01. Define the following terms:

ANS: Classification:

Biological classification is the process by which scientists group living organisms. Organisms are classified based on how similar they are. Historically, similarity was determined by examining the physical characteristics of an organism but modern classification uses a variety of techniques including genetic analysis.

Taxon:

In biology, a taxon is a group of one or more populations of an organism or organisms seen by taxonomists to form a unit.

Nomenclature:

The devising or choosing of names for things, especially in a science or other discipline.

Pure culture:

A culture in which only one strain or clone is present

Identification:

The action or process of identifying someone or something or the fact of being identified.

Mol%G+C:

DNA is made up of base pairs: guanine –cytosine and adenine –thymine. Of the total number of nucleotide bases present in DNA, that percentage represented by guanine plus cytosine is termed the mol% G+C.

Berge's Manual

Berge's Manual of Systematic Bacteriology is the main resource for determining the identity of prokaryotic organisms, emphasizing bacterial species, using every characterizing aspect.

Bacterial strain: A strain is a genetic variant or subtype of a micro-organism (e.g., virus or bacterium or fungus). For example, a "flu strain" is a certain biological form of the influenza or "flu" virus.

In an effort to be more objective about grouping bacteria, a scientist may determine many characteristics (usually 100 to 200) for each strain studies, giving each characteristic equal weight. Then using a computer, he or she calculate the similarity (% S) of each strain to every other strain. For any two strains, this is:

$$\%S = \frac{NS}{NS + ND}$$

Q. 02. Why is it essential to classify microorganisms and what must be done before they can be classified?

ANS: Classification is important because it allows scientists to identify, group, and properly name organisms via a standardized system (Linnaeus Taxonomy) based on similarities found in the organisms. Knowledge of DNA/RNA (genetics), adaptations (Evolution), and embryonic development (Embryology) of other known organisms helps to better understand the new organism as a whole. In order to classify microorganism, it is important to learn their characteristics.

Q. 03. List the major kinds of microbial characteristics and indicates those that must be determined by examination of individual cells.

ANS: The major characteristics of microorganisms are as follows:

- Morphological characteristics
- Chemical characteristics
- Cultural characteristics
- Metabolic characteristics
- Antigenic characteristics

- Genetic characteristics
- Pathogenicity
- Ecological characteristics

The morphological characteristics are determined by the examination of individual cells.

Q. 04. Why the type strain is the most important strain in a bacterial species?

ANS: The type strain is the strain that is designated to be the permanent reference specimen for the species. It is the strain to which all other strains must be compared to see if they resemble it closely enough to belong to the species. Therefore, type strains are particularly important.

Q. 05. Explain the subjectivity that exists in the definition of a bacterial species?

ANS: In the definition of bacterial species, the phrase "considered sufficiently similar to the type strain" indicates that the definition contains an element of subjectivity. In other words, the criteria which one taxonomist believes to constitute "sufficiently similar" may be quite different from those used by another taxonomist.

Q. 06. (a) If two microorganisms have an identical mol% G+C value for their DNA are they necessarily related?

(b) If two microorganisms have very different mol% G+C values for their DNA are they necessarily unrelated?

ANS: The most reliable method of classification is based on the degree of genetic relatedness between organisms. This method based on the most fundamental aspect of organisms, their hereditary material. In the 1960s the development of that branch of science known as molecular biology provided techniques by which the DNA of one organism could be compared with that of other organisms. At first only crude comparisons could be made, based on mol% G+C values. It is true that two organisms of the same species that are very closely related will have very similar mol% G+C values, and it is also true that two organisms having quite different mol% G+C values are not very closely related. However, it is also important to realize that organisms that are completely unrelated may have similar mol% G+C values.

Chapter No.05

The Morphology and Fine Structure of Bacteria

QUESTIONS:

Q. 01. How does the cell's surface area/volume ratio compare with that of larger organisms? What advantage does a high surface area/volume ratio offer? What constraints does it place on a cell?

ANS: Large organisms have a smaller surface area to the volume ratio. Due to the small surface area, gaseous exchange cannot be carried out efficiently as the distance between surface area and the respiring cells in the organism are further away (since the volume is big.). This leads to the decrease in the rate of diffusion, since it has to travel quite a distance.

A relatively large surface through which nutrients can enter (waste product leave) compared to a small volume of cell substance to be nourished accounts for the unusually high rate of growth and metabolism of bacteria.

Q. 02. What bacteria cell's structure may help to increase the cell's surface area/Volume ratio?

ANS: Bacteria cell is a single cell so as the size of a cell becomes smaller, the ratio of surface area (SA) to volume (V) increases. This is because SA is responsible for the transportation of nutrients into the cell and waste out of the cell. The inside of the cell uses up the materials and so need a larger SA to keep up with the waste excretion so that it doesn't accumulate.

Q. 03. If you performed a microscope examination of an appropriately stained preparation of *Staphylococcus aureus* would you expect all the cells to be arranged in clusters?

Explain.

ANS: If we prepare a staphylococcus aureus stain to examine on a microscope, we will see grape-like cluster.

Q. 04. Explain why some species of cocci appear as chains but others appear in a cuboidal arrangement.

ANS: Cocci appear in several characteristic arrangements, depending on the plane of cellular division and whether the daughter cells stay together following cell division. Cells of some species of cocci divide in one plane and remain attached to form chains.

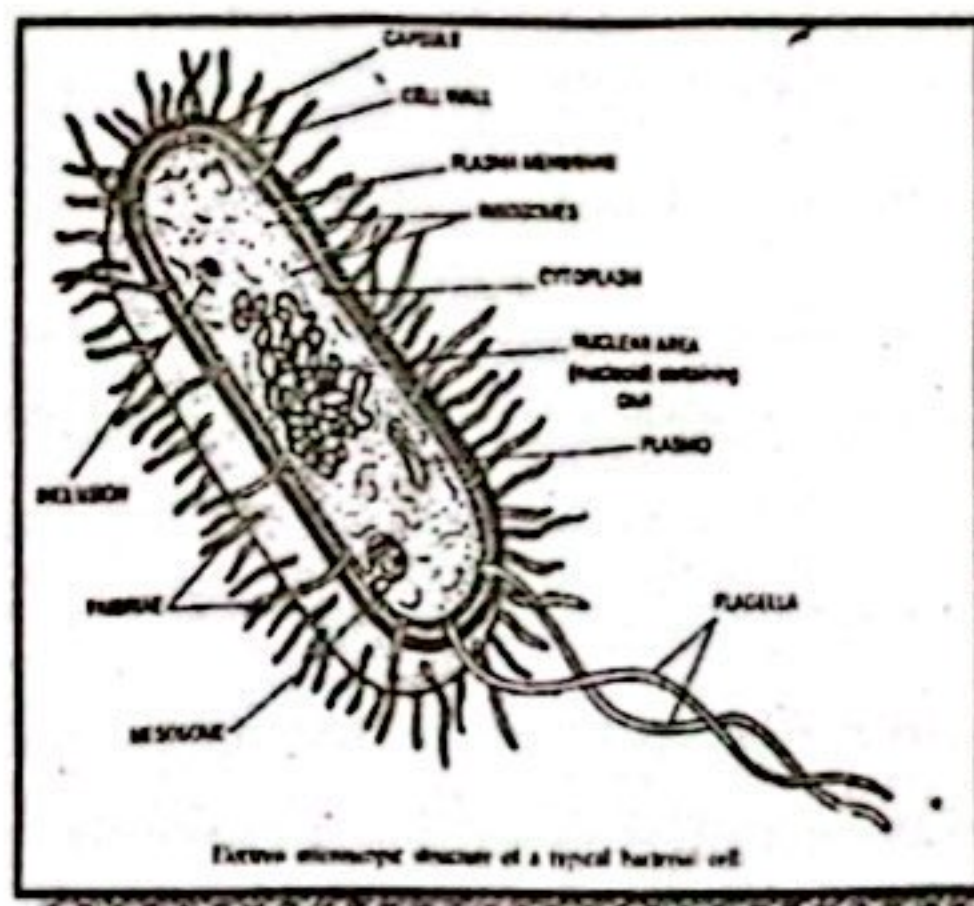
Example: Streptococci

Whereas as some species divide in three planes, in a regular pattern, producing a cuboidal arrangement of the cells.

Example: Sarcinae.

Q. 05. Draw a typical bacterial cell and identify all parts.

ANS:



Q. 06. Contrast propulsion by a bacterial flagellum with that of a screw propeller on a submarine.

ANS: Bacteria propel themselves by rotating their helical flagella. The principle involved can be illustrated by imagining the penetration of a piece of cork by a corkscrew. If one tries to ram the corkscrew directly through the cork, great force will probably be needed. On the other hand, if one merely rotates the corkscrew, the cork can be easily penetrated. In the case of bacteria, the cork is

analogous to the viscous medium and the corkscrew to the helical flagellum. It is apparent from this analogy that a mutant bacterium having straight rather than helical flagellum would be unable to swim. It is known that the flagella motor is driven by the proton motive force, i.e., the force driven from the electrical potential and the hydrogen-ion gradient across the cytoplasmic membrane. Recent studies suggest that the concentration of cGMP within the cell governs the direction in which the rotation occurs.

Q. 07. What functions chemo taxis, photo taxis and magneto taxis have for bacteria in their natural habitats?

ANS: Chemotaxis is the movement of an organism in response to a chemical stimulus. Bacteria, and other single-cell or multicellular organisms direct their movements according to certain chemicals in their environment. This is important for bacteria to find food (e.g., glucose) by swimming toward the highest concentration of food molecules, or to flee from poisons (e.g., phenol).

Magneto taxis describes an ability to sense a magnetic field and coordinate movement in response. Motile aquatic bacteria orient to the earth's magnetic field even after death, without biologically sensing the field. They are now called simply *magnetic bacteria*. These bacteria (e.g. *Magneto spirillum*, *Magneto tacticum*) contain internal structures known as magnetosomes. They appear as a chain of dark, membrane-bound crystals - often magnetite (Fe_3O_4). By orienting toward the earth's poles, marine bacteria are able to direct their movement downwards, towards the sediment and even towards the oxygen-deficient area more favorable for growth. However, these bacteria are found even at the earth's magnetic equator, where the field is directed horizontally.

Phototaxis is a kind of taxis or locomotory movement that occurs when a whole organism moves towards or away from stimulus of light. Phototaxis is called positive if the movement is in the direction of increasing light intensity and negative if the direction is opposite. Positive phototaxis occurs when a bacterium swims by chance out of the area illuminated by the microscope. Entering darkness signals the cell to reverse flagella rotation direction and reenter the light. The second type of phototaxis is true phototaxis, which is a directed movement up a gradient to an increasing amount of light. This is analogous to positive chemotaxis except that the attractant is light rather than a chemical.

Q. 08. What problems associated with the shape and motility of Spiro plasmas still remain to be solved?

ANS: Spiro plasmas are gram-positive bacteria. They lack cell walls and flagella, but have internal cytoskeletons and are motile and chemotactic. The Spiro plasmas are unique in having well-defined basic helical cell geometry. A flat cytoskeletal ribbon of parallel fibrils is attached to the inside of the cellular tube. Both tube and cytoskeleton are mutually coiled into a dynamic helix driven by differential length changes of the fibrils, which function as linear motors.

Q. 09. What functions might a capsule serve for the following bacteria?

ANS:

a) A pathogenic bacterium

The capsule is considered a virulence factor because it enhances the ability of bacteria to cause disease (e.g. prevents phagocytosis). The capsule can protect cells from engulfment by eukaryotic cells, such as macrophages.

b) A soil bacterium where the soil is periodically subjected to drought conditions Capsules also contain water which protects the bacteria against desiccation. They have water containing binding molecules which serve in the dry conditions.

c) A bacterium living in a flowing stream.

It has the capsule that promotes the attachment of bacterium to the surfaces in water.

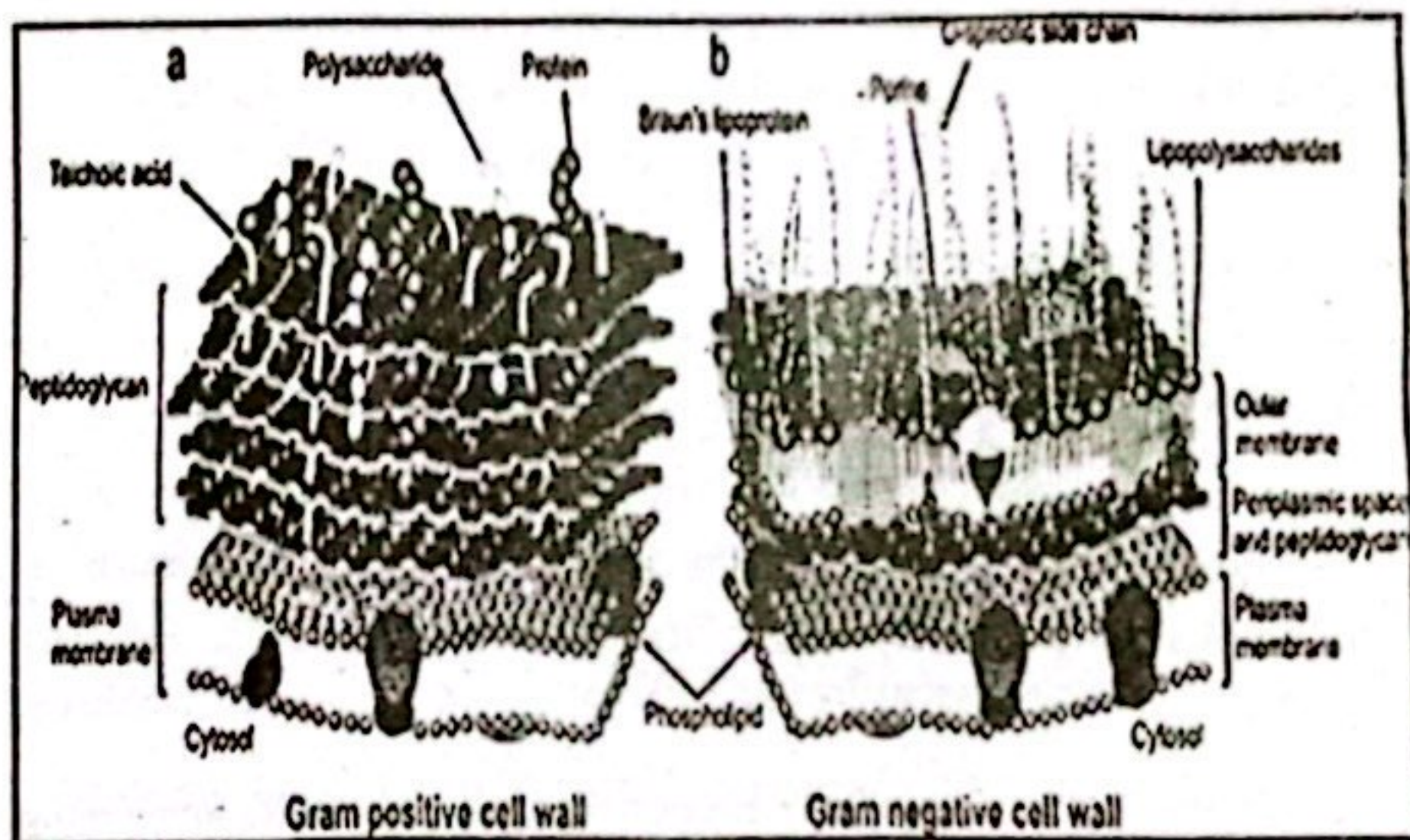
Q. 10. Why are Gram-negative eubacteria usually much easier to disrupt by sonic oscillation than Gram-positive eubacteria?

ANS: Gram-positive bacteria usually have a much greater amount of peptidoglycan in their cell walls than do Gram-negative bacteria; it may account for 50 percent or more of the dry weight of the wall of some Gram-positive species, but only about 10 percent of the wall of Gram-negative bacteria. Therefore, Gram-negative bacteria are much easier to disrupt by sonic oscillation.

Q. 11. Compare the structure and chemistry of cell walls of Gram-positive eubacteria versus those of Gram-negative eubacteria. List some major differences between the cell walls of archaeo bacteria versus those of eubacteria.

ANS: Cell walls of Gram-positive eubacteria versus those of Gram-negative eubacteria

- **Gram-positive bacteria** usually have a much greater amount of peptidoglycan in their cell walls than do Gram-negative bacteria; it may account for 50 percent or more of the dry weight of the wall of some Gram-positive species, but only about 10 percent of the wall of Gram-negative bacteria.
- The walls of some Gram-positive bacteria like *Streptococcus pyogenes* contain polysaccharides covalently linked to peptidoglycan.
- The walls of some Gram-positive bacteria have teichoic acid.
- Gram-positive bacteria contain very little lipids.
- The walls of **Gram-negative bacteria** are more complex than those of Gram-positive bacteria. The difference is the presence of an **outer membrane** that surrounds a thin underlying layer of peptidoglycan.
- The walls of Gram-negative bacteria are rich in lipids in contrast to those of Gram-positive bacteria.
- Outer membrane of the wall of Gram-negative bacteria is consisting of **phospholipids, proteins and lipopolysaccharides**.
- Outer membrane of Gram-negative bacteria also contains **porins**, which have essential role in permeability of outer membrane.



FEATURES	GRAM NEGATIVE BACTERIA	GRAM POSITIVE BACTERIA
Peptidoglycan layer	Thin (single-layered)	Thick (multilayered)
Teichoic acids	Absent	Present in many
Periplasmic space	Present	Absent
Outer membrane	Present	Absent
Lipopolysaccharide (LPS) content	High	Virtually none
Lipid and lipoprotein content	High (due to presence of outer membrane)	Low (acid-fast bacteria have lipids linked to peptidoglycan)

Differences between the cell walls of archaeobacteria versus those of eubacteria

Cell walls of archaeobacteria do not contain peptidoglycan. Their walls are usually composed of proteins, glycoproteins, or polysaccharides while the walls of eubacteria have peptidoglycan. This is the main difference between the walls of the archaeobacteria and the eubacteria.

Q. 12. What function do the porins of the outer membrane of Gram-negative bacteria cell wall serve? What function do cytoplasmic membrane/outer membrane adhesions serve? ANS
 Porins are beta barrel proteins that cross a cellular membrane and act as a pore, through which molecules can diffuse. Porins are large enough to allow passive diffusion, i.e., they act as channels that are specific to different types of molecules. They are present in the outer membrane of gram-negative bacteria and some gram-positive bacteria of the group Mycolata, the mitochondria, and the chloroplast.

➤ To transport medium-sized or charged molecules across, the molecules move through a porin, a water-filled channel or pore. Porins typically control the diffusion of small metabolites like sugars, ions, and amino acids. In gram-negative bacteria, the inner membrane is the major permeability

barrier, whereas the outer membrane contains porins, which render it largely permeable to molecules less than about 1500 Daltons.

➤ Porins are chemically selective – transporting only one group of molecules – or may be specific for one molecule. Beta-lactam and fluoroquinolone antibiotics must pass through porins to reach their targets in gram negative bacteria.

➤ Bacteria can develop resistance to these antibiotics by mutating the gene that encodes the porin – the antibiotics are then excluded from passing through the outer membrane.

Q. 13. In what kind of bacterial cell structures would be most likely to find the following compounds:

ANS:

a. Peptidoglycan

In the cell walls of Gram-positive and Gram-negative eubacteria.

Teichoic acid

In the walls of the *Staphylococcus aureus* and *Streptococcus faecalis*.

b. Calcium dipicolinate

In the endospores of the bacteria.

c. Cholesterol

In the cytoplasmic membrane of bacteria.

d. Lipopolysaccharides

In the outer membrane of the Gram-negative bacteria.

e. Phytanols ether-linked to glycerol

In the cytoplasmic membrane of the archaeobacteria.

Q. 14. Is spore formation in bacteria a method of reproduction or a means of multiplication?

ANS: Spore formation is the process to produce a spore when the bacteria is starving or sense change in the environment, like extreme temperatures and drought, This spore is a protective, thick cell walls and can help the bacteria survive for several years by living in a dormant state.

Spore formation is the means of reproduction in the bacteria.

Q. 15. What are the similarities and differences between protoplasts and spheroplasts?

ANS: Difference:

The term protoplast refers to the spherical shape assumed by Gram-positive bacteria. Spheroplast refers to the spherical shape assumed by Gram-negative bacteria. The difference is essentially the presence of a single membrane, in the case of the protoplast, and the two membranes (inner and outer) of the Gram-negative spheroplasts. It is also possible to generate a gram-negative protoplast by the removal of the outer membrane. Thus, in essence, protoplast refers to a bacterial sphere that is bounded by a single membrane and spheroplast refers to a sphere that is bounded by two membranes.

Similarities:

Both are spherical shaped; both are prepared by similar method by treating with enzyme.

Spheroplasts are osmotically fragile, and will lyse if transferred to a hypotonic solution.

Q. 16. Is it proper to refer to bacterial cells as containing a typical nucleus?

ANS: No, most bacterial cells do not have nuclei, because bacterial cells are prokaryotes. (One possible exception is in the Planctomycetes, one species, *Gemmata obscuriglobus*, may have a nucleus-like structure. Because there is no nucleus, transcription and translation can be coupled as a gene is being transcribed, ribosomes can immediately begin the process of translation. This obviously affects how genes are regulated and has implications for horizontal gene transfer and recombination as well as cell division and many other basic biological processes.

Q. 17. Name several cytoplasmic inclusions or substances. What function might be associated with each of these?

ANS: Inclusions are considered to be nonliving components of the cell that do not possess metabolic activity and are not bounded by membranes. The most common cytoplasmic inclusions of complex inorganic polyphosphate. These granules are called metachromatic granules due to their displaying the metachromatic effect; they appear red or blue when stained with the blue dyes methylene blue or toluidine blue. Some bacteria that live in aquatic habitats form gas vacuoles that provide buoyancy.



Chapter No.06

The Cultivation of Bacteria

QUESTIONS:

Q. 01. What nutritional requirements in terms of chemical are needed by all forms of life for growth and cellular maintenance?

ANS: All forms of life require following substance:

Essential Nutrients: Two categories of essential nutrients; macro-nutrients and micro-nutrients.

- Macro-nutrients are needed in larger amounts.
- Used to help with cell structure and the cell's metabolism.
- Examples are proteins, and carbohydrates.
- Micro-nutrients or trace elements are needed in a lot smaller amount.
- They help enzyme function and help to maintain protein structure.
- They include elements such as zinc, manganese, and nickel.

The source of common essential nutrients is: CHNOPS;

- Carbon
- Hydrogen
- Nitrogen
- Oxygen
- Phosphorous
- Sulfur
- Carbon Content Nutrients: Carbon content is another way that nutrients are categorized.

Sources of nutrients are extremely varied and some microbes will obtain their nutrients entirely from inorganic sources and others require a combination of organic and inorganic sources.

Oxygen Sources: Because oxygen is a major component of organic compounds such as carbohydrates, lipids, nucleic acids, and proteins, it plays an important role in the structural and enzymatic functions of the cell. Oxygen is likewise a common component of inorganic salts such as sulfates, phosphates, nitrates, and water. Free gaseous oxygen (O_2) makes up 20% of the atmosphere. It is absolutely essential to the metabolism of many organisms.

Hydrogen Sources: Major element in all organic and several inorganic compounds such as water, salts, hydrogen, sulfide, methane, and hydrogen gas. (Used/produced by microbes) **Essential** in biochemistry of cells: form hydrogen bonds (oxygen/nitrogen), free energy in reactions of respiration, maintain pH of solutions.

Q. 02. Distinguish between (a) phototrophs and chemotrophs (b) lithotrophs and organotrophs, (c) autotrophs and heterotrophs.

ANS: (a)

Phototrophs	Chemotrophs
Phototrophs are the organisms that carry out photon capture to acquire energy. They use the energy from light to carry out various cellular metabolic processes. It is a common misconception that phototrophs are obligatorily photosynthetic. Many, but not all, phototrophs often photosynthesize: they convert carbon dioxide into organic material anabolically to be utilized structurally, functionally, or as a source for later catabolic processes (e.g. in the form of starches, sugars and fats). All phototrophs either use electron transport chains or direct proton pumping to establish an electrochemical gradient which is utilized by ATP synthase, to provide the molecular energy currency for the cell. Phototrophs can be either autotrophs or heterotrophs.	Chemotrophs are organisms that obtain energy by the oxidation of electron donors in their environments. These molecules can be organic (chemoorganotrophs) or inorganic (chemolithotrophs). The chemotroph designation is in contrast to phototrophs, which utilize solar energy. Chemotrophs can be either autotrophic or heterotrophic. Chemotrophs are commonly found in ocean floors where sunlight cannot reach them because they are not dependent on solar energy. Ocean floors often contain underwater volcanos that can provide heat to substitute sunlight for warmth.

(b)

Lithotrophs	Organotrophs
Lithotrophs are a diverse group of organisms that use inorganic substrate (usually of mineral origin) to obtain reducing equivalents for use in biosynthesis (e.g., carbon dioxide fixation) or energy conservation (i.e., ATP production) via aerobic or anaerobic respiration known as chemolithotrophs are exclusively microorganisms; no known macrofauna possesses the ability to utilize inorganic compounds as energy sources. An example of this is chemo lithotrophic bacteria.	An organotroph is an organism that obtains hydrogen or electrons from organic substrates. This term is used in microbiology to classify and describe organisms based on how they obtain electrons for their respiration processes. Some organotrophs such as animals and many bacteria, are also heterotrophs. Organotrophs can be either anaerobic or aerobic.

(c)

Autotrophs	Heterotrophs
An autotroph ("self-feeding", from the Greek <i>autos</i> "self" and <i>trophe</i> "nourishing") or producer, is an organism that produces complex organic compounds (such as carbohydrates, fats and proteins) from simple substances present in its surroundings, generally using energy from light (photosynthesis) or inorganic chemical reactions (chemosynthesis). Autotrophs can be photoautotrophs or chemoautotrophs.	A heterotroph is an organism that cannot fix carbon from inorganic sources (such as carbon dioxide) but uses organic carbon for growth. Heterotrophs can be further divided based on how they obtain energy; if the heterotroph uses light for energy, then it is considered a photoheterotroph, while if the heterotroph uses chemical energy, it is considered a chemoheterotroph.

Q. 03. Suppose you want to determine the occurrence of the vitamin biotin in batch of milk. *Leuconostoc mesenteroides* require biotin for growth. Using this bacterium and chemically defined medium to which a small sample of sterilized milk can be added devise an experiment that could indicate whether biotin is present in milk?

ANS: Vitamin H, more commonly known as biotin, is part of the B complex group of vitamins. All B vitamins help the body to convert food (carbohydrates) into fuel (glucose), which is used to produce energy. These B vitamins, often referred to as B complex vitamins, also help the body metabolize fats and protein. B complex vitamins are needed for healthy skin, hair, eyes and liver. They also help the nervous system function properly. Dairy products such as milk and cheese are natural sources of this vitamin. Cheddar cheese provides 0.4 to 2 micrograms of biotin per ounce serving.

Q. 04. What condition of cultivation would allow you to grow selectively?

ANS: (a) *Neisseria gonorrhoeae* from a clinical specimen:

Urethral specimens: Discharge from the meat us is preferred for the detection of *N gonorrhoeae*. Intraurethral swab can be used for the detection of gonococci. To increase the chance of detecting the organisms, swab samples should be collected from patients who have not voided for at least 2 h. The swabs are suitable for smear preparation, culturing on appropriate media or for transport to other laboratories.

Urine: Urine is one of the specimen types suitable for nucleic acid tests for diagnosing *N gonorrhoeae* infections in males and females. Leak-proof containers should be provided to patients for the collection of urine specimens.

(b) an extreme halophile from a sample of sea salt:

Of particular note are the extreme halophiles or haloarchaea (often known as halobacteria), a group of archaea, which require at least a 2 M salt concentration and are usually found in saturated solutions (about 36% w/v salts). These are the primary inhabitants of salt lakes, inland seas, and evaporating ponds of seawater, where they tint the water column and sediments bright colors. These species most likely perish if they are exposed to anything other than a very high-concentration, salt-conditioned environment. These prokaryotes require salt for growth. The high concentration of sodium chloride in their environment limits the availability of oxygen for respiration. Their cellular machinery is adapted high salt concentrations by having charged amino acids on their surfaces, allowing the retention of

water molecules around these components. They are heterotrophs that normally respire by aerobic means. Most halophiles are unable to survive outside their high-salt native environments. Indeed, many cells are so fragile that when placed in distilled water, they immediately lyse from the change in osmotic conditions.

(c) A nitrogen fixing bacterium from a soil sample:

Nitrogen-fixing bacteria grown in semisolid malate agar were isolated from the soils and various plant roots of cultivated and non-cultivated sites in and near Tokyo University Forest in Aichi, and characteristics of the isolates were compared. All of these isolates reduced nitrate and a large portion of them formed gas in anaerobic culture. The incidence of nitrogen-fixing bacteria from the forest site was much lower and their presence was demonstrated in about 40% of the forest samples. The characteristics of the forest isolates were generally different from those of *A. brasilense*, as they required yeast extract for nitrogen-fixing growth. More than half of the forest isolates failed to reduce nitrate and none of them produced gas from nitrate.

(d) *Thiobacillus thiooxidans* from mixed culture of bacteria.

Thiobacillus Agar is used for the isolation and maintenance of *Thiobacillus* species. The medium contains three inorganic sulphates and a thiosulphate. Phosphate serves as a buffer while sodium chloride maintains the osmotic balance of the medium. Samples are inoculated into Thiobacillus Broth. After incubation at 25-30°C for about 7 days or more, turbidity or sulphur precipitation on the surface of the liquid or against the walls of the flasks, indicates growth of bacteria. Isolation is subsequently done on Thiobacillus Agar. *Thiobacillus* forms small sulphur impregnated colonies with clear zones, indicating acid formation from thiosulphate oxidation.

Q. 05. Most medically important anaerobic bacteria are non-stringent anaerobes, but some are the stringent type. Describe two methods recommended for cultivating the latter organisms?

ANS: The Anaerobic jar: Undoubtedly, the best general method of anaerobic cultivation is the use of the anaerobic jar. It is capable of producing conditions sufficiently anaerobic to grow the strictest anaerobes (e.g. *Clostridium tetani*) and allows specimens to be directly streaked out, in the normal way, on solid media.

The BTL Anaerobic jar: A seamless metal or plastic vessel, sufficiently large to contain Petri dishes and tubes is fitted with a gas-tight lid. The lid has inlet and outlet valves and, on its inner surface,

room-temperature catalyst enclosed within a wire gauze envelope. The catalyst consists of alumina pellets coated with 0.5 per cent palladium which catalyses the combination of oxygen in the jar with hydrogen to form water.

Method of use:

1. Place cultures in the jar, taking care to loosen the tops of, screw-capped bottles. Secure the lid.
2. Close the inlet valve, connect the outlet valve to a vacuum pump, and evacuate to approximately 10 cm of mercury. Close the outlet valve.
3. Connect the inlet valve to a supply of purified hydrogen gas. A convenient method is to supply the gas from a football bladder or an-aesthetic apparatus bag that has been filled from a cylinder.
4. Open the inlet valve. The hydrogen will quickly enter the jar and the residual oxygen will be burnt to form water on the gauze cover of the catalyst.
5. After 3 min the reaction should be complete and the inlet valve may be closed, the bag disconnected and the jar placed in an ordinary incubator.

For optimum growth of strict anaerobes, 5-10 per cent CO₂ is often incorporated in the hydrogen supply. One advantage of this type of jar is that the process of evacuation causes oxygen dissolved in liquid media to be displaced by the reduced pressure, thereby ensuring correct anaerobic conditions within the tubes.

THE 'GAS PAK' SYSTEM: The principle employed here is the generation of both hydrogen and carbon dioxide by adding water to pre-packed chemicals, thereby eliminating the need for any external apparatus and manipulations.

Q. 06. Is nutrient broth a universal medium (that is, can support the growth of every kind of bacteria)? Explain.

ANS: Hardy Diagnostics Nutrient Broth is a general-purpose growth medium recommended for use in the cultivation of non-fastidious microorganisms. Nutrient Broth can be used as a pre enrichment medium and may also be used in accordance with many standard methods procedures for testing food, milk and dairy products.

Hardy Diagnostics Nutrient Broth is a liquid medium that contains peptone and beef extract. Nutrients necessary for the replication and growth of a large number of non-fastidious microorganisms are provided by this simple formulation. Water soluble substances including carbohydrates, vitamins, organic nitrogen compounds and salts are present in beef extract. Pancreatic digest of gelatin supplies the principle source of organic nitrogen in the form of amino acids and long-chain fatty acids.

Q. 07. If you wanted to buffer a culture medium to keep the pH at 5.5, you would try to use a buffer having what pK_a ? Why?

ANS: Radical shifts in pH can be prevented by incorporating a buffer into medium. A buffer is mixture of a weak acid and its conjugate base. Such mixtures have maximum buffering capacity at the pH where the concentrations of the acid equals that of its conjugate base. The pH value is called pK_a and its negative logarithm of the dissociation constant of the acid. The extent to which a medium should or may be buffered depends on its intended purpose and is limited by the buffering capacity of the compounds used. Some large fermentation apparatuses are equipped with automatic controls that maintain a constant pH.

Q. 08. List some of the ways whereby you could enhance the growth of an aerobic bacterium.

ANS:

- Nutrition concentration
- Temperature
- pH
- Available Water

Q. 09. Under what circumstances might you wish to use silica gel as a solidifying agent for a culture medium?

ANS:

- Silica gel is used as an inorganic solidifying agent for autotrophic bacteria.
- Silica gel is used at conc. of 1.5 to 2.0 that is not degraded by most bacteria.
- Silica gel has soft custard like consistency used to determine the motility of bacteria.

- Silica gel is used to determine the characteristics of colonies.

Q. 10. Indicate the various toxic derivatives of oxygen and explain how aerobic organisms might protect themselves against these derivatives?

ANS:

Various toxic derivatives of oxygen are:

- Superoxide radicals
- Hydrogen peroxide
- Hydroxyl radicals
- Singlet oxygen

Protective mechanisms:

- Some aerobic and facultative organisms have enzyme known as superoxide dismutase.
- Some bacteria have enzymes that are catalase and peroxidase.



Microbial Agents of Disease: Viruses

QUESTIONS:

Q. 1. For immunization against polio give advantages and disadvantages of (a) The Salk vaccine (b) The Sabin vaccine.

ANS: The Salk killed vaccine, given by injection, is the approved vaccine in the United States today because of its greater safety. The Sabin oral, attenuated vaccine is given in many parts of the world, particularly in underdeveloped countries, because of its ease of administration. **Q. 2. Why is it impractical to prepare a vaccine against the common cold?**

ANS: Peramivir injection helps shorten the time that flu symptoms such as a stuffy or runny nose, sore throat, cough, muscle or joint aches, tiredness, headache, fever, and chills last. There are various subtypes of influenza virus so vaccine prepared for one strain is ineffective for another.

Q. 3. Explain how secretory IgA can act to prevent infection by some specific viruses.

ANS: Immunoglobulin A (IgA) is the first line of defense in the resistance against infection, via inhibiting bacterial and viral adhesion to epithelial cells and by neutralisation of bacterial toxins and virus, both extra- and intracellularly. Secretory IgA antibodies in mucosal secretions contribute to immune homeostasis by limiting access of microbial and environmental antigens to the body proper, maintaining the integrity of the epithelial barrier, and shaping the composition of the commensal microbiota.

Q. 4. What role might recombinant DNA techniques play in the development of vaccine against viral diseases? Give example.

ANS: The recombinant plasmids containing a foreign gene are purified from the bacteria, and the "naked" DNA is injected directly into the animal, usually intramuscularly or intradermal (into the skin). The animal's cells take up the DNA, and an immune response is induced to the protein expressed from the foreign gene. The classical example of recombinant protein vaccines currently in use in humans is the vaccine against hepatitis B.

Q. 5. List some antiviral drugs and explain their mechanism of action.

ANS:

7. MECHANISM OF SELECTED ANTIVIRAL AGENTS

Lamivudine	Inhibition of viral DNA polymerase and reverse transcriptase	Hepatitis B (chronic cases), human immunodeficiency virus type 1
Oseltamivir	Inhibition of viral neuramidase	Influenza A
Penciclovir	Metabolized to penciclovir triphosphate, which inhibits viral DNA polymerase	Herpes simplex
Ribavirin	Interference with viral messenger RNA	Lassa fever, hantavirus (hemorrhagic fever renal syndrome), hepatitis C (in chronic cases in combination with interferon-α) RSV in children and infants
Amantadine	Blockage of the M2 protein ion channel and its ability to modulate intracellular pH	Influenza A
Valacyclovir	Same as acyclovir	Herpes simplex, varicella-zoster, cytomegalovirus
Vidarabine	Inhibits viral DNA synthesis	HSV-1, HSV-2, and VZV; its use is limited to treatment of immunocompromised patients with HSV keratitis
Zanamivir	Inhibition of viral neuramidase	Influenza A

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Q. 6. Indicate the features of small pox that made eradication of this disease possible. Give the essential features of the grand strategy that was used for the eradication of this disease.

ANS: Several characteristics of smallpox led to its control and eventual eradication: 1) cases could be identified because smallpox is a clinically evident disease and there is no subclinical illness; 2) the disease moves relatively slowly— transmission does not occur during prodrome and maximum transmission is at the time. There were many factors that uniquely favored smallpox eradication: No other disease has features that made diagnosis and surveillance for infection so easy. Because every infected person had a characteristic rash, the presence or absence of the virus could be determined quickly in every geographic area.

Q. 7. Rhinoviruses and coronaviruses can cause the common cold. How do these two kinds of viruses easily be eradicated?

ANS: Rhinovirus infections are predominantly mild and self-limited; thus treatment is generally focused on symptomatic relief and prevention of person to person spread and complications. The main steps of therapy include rest, hydration, first generation antihistamines and nasal decongestants. Like Rhinoviruses, human coronaviruses represent causative agents of the common cold. At least three immunological groups exist. Vaccines have been made to prevent the spread of Covid 19.

Q. 8. Could the following viruses easily be eradicated?

ANS: (a) Toga viruses: Toga viruses are important veterinary pathogens, and are transmitted via mosquitoes. Some of them are zoonotic viruses, including Venezuelan equine encephalitis virus (VEEV). Rubella virus is the only member of togavirus family that causes significant disease in human—German measles. Two infectious diseases have successfully been eradicated: smallpox and rinderpest. There are also four ongoing programs, targeting poliomyelitis, yaws, dracunculiasis, and malaria.

(b) Orthomyxoviruses

The orthomyxoviruses (influenza viruses) constitute the genus Orthomyxovirus, which consists of three types (species): A, B, and C. These viruses cause influenza, an acute respiratory disease with prominent systemic symptoms.

(c) Rabies viruses

Dogs are the main source of human rabies deaths and contribute up to 99% of all rabies transmissions to humans; however, rabies eradication is possible through vaccination of dogs and the prevention of dog bites, according to the health agency. The rabies virus is a very fragile virus. As soon as the saliva dries, the virus is no longer infectious. The virus is easily killed by soaps, detergents, bleach, alcohol and ultraviolet light.

(d) Adenoviruses

There is no specific treatment for people with adenovirus infection. Most

adenovirus infections are mild and may require only care to help relieve symptoms, such as over-the-counter pain medicines or fever reducers. Always read the label and use medications as directed.

Complete eradication is not possible. There are still cases reported for these viruses.

Q. 9. What recent development in rabies immunization have led to a safer yet effective vaccine.

ANS: Louis Pasteur developed the earliest effective vaccine against rabies that was first used to treat a human bite victim on 6 July 1885. An animal can be considered immunized within 28 days after initial vaccination, when a peak rabies virus antibody titer is reached. The development of molecular adjuvants that induce faster, stronger immune responses with less antigen has yielded exciting preclinical results and appears to edge us closer to a better rabies vaccine.

Q. 10. Why is a live vaccine rather than an inactivated vaccine required for immunization against small pox?

ANS: Live-attenuated vaccines: Because these vaccines are so similar to the natural infection that they help prevent, they create a strong and long-lasting immune response. Just 1 or 2 doses of most live vaccines can give you a lifetime of protection against a germ and the disease it causes. Live virus vaccines use the weakened (attenuated) form of the virus. The measles, mumps, and rubella (MMR) vaccine and the varicella (chickenpox) vaccine are examples. Killed (inactivated) vaccines are made from a protein or other small pieces taken from a virus or bacteria.

Pathogens for inactivated vaccines are grown under controlled conditions and are killed as a means to reduce infectivity (virulence) and thus prevent infection from the vaccine. The virus is killed using a method such as heat or formaldehyde.

Q. 11. Contrast infection caused by herpes simplex virus type 1 (HSV-1) with those caused by HSV-2.

ANS: HSV-1 is mainly transmitted by oral-to-oral contact to cause oral herpes (which can include symptoms known as "cold sores"), but can also cause genital herpes. HSV-2 is a sexually transmitted infection that causes genital herpes. Both HSV-1 and HSV-2 infections are lifelong.

Q. 12. Contrast hepatitis virus type A with hepatitis virus type B.

ANS: "Hepatitis A virus can cause acute liver disease, but can heal within a few months. It can cause high spiking fevers and is more severe in adults than in children," says Gulati. "Hepatitis B virus has an 85 percent recovery rate, while 15 percent develop cirrhosis or cancer of the liver." Hepatitis B is a blood-borne pathogen; its primary mode of transmission is through direct blood-to-blood contact

with an infected person. In contrast, hepatitis A can be spread by fecal-oral transmission or by consuming food or water that has been contaminated.

Hepatitis E, also called enteric hepatitis (enteric means related to the intestines), is similar to hepatitis A, and more prevalent in Asia and Africa. It is also transmitted through the fecal-oral route. It is generally not fatal, though it is more serious in women during pregnancy and can cause fetal complications.

Q. 13. Briefly define the following items:

ANS: Negri bodies:

Negri bodies are round or oval inclusions within the cytoplasm of nerve cells of animals infected with rabies. Negri bodies may vary in size from 0.25 to 27 μm . They are found most frequently in the pyramidal cells of Ammon's horn, and the Purkinje cells of the cerebellum.

Guarnieri bodies

Guarnieri bodies - intracytoplasmic acidophilic inclusion bodies observed in epithelial cells in smallpox and vaccinia infections, and which include aggregations of Paschen bodies or virus particles. Guarnieri gelatin agar - a type of agar used for the cultivation of *Streptococcus pneumoniae*

HBs Ag

Laboratory diagnosis of hepatitis B infection focuses on the detection of the hepatitis B surface antigen HBs Ag. WHO recommends that all blood donations be tested for hepatitis B to ensure blood safety and avoid accidental transmission to people who receive blood products. A positive HBs Ag test result means that you are infected and can spread the hepatitis B virus to others through your blood. anti-HBs or HBs Ab (Hepatitis B surface antibody) - A "positive" or "reactive" anti-HBs (or HBs Ab) test result indicates that a person is protected against the hepatitis B virus.

Adenovirus-associated virus

Adenoviruses are common viruses that cause a range of illness. They can cause cold-like symptoms, fever, sore throat, bronchitis, pneumonia, diarrhea, and pink eye (conjunctivitis). You can get an adenovirus infection at any age. When a latently infected cell is super-infected with a helper virus, such as adenovirus or herpes simplex virus, the AAV gene expression program is activated leading to

the AAV Rep-mediated rescue (i.e., excision) of the provirus DNA from the host cell chromosome followed by replication and packaging of the viral genome.

Reverse transcriptase

Reverse transcriptase (RT), also known as RNA-dependent DNA polymerase, is a DNA polymerase enzyme that transcribes single-stranded RNA into DNA. This enzyme is able to synthesize a double helix DNA once the RNA has been reverse transcribed in a first step into a single-strand DNA.

Latent infection

A latent infection is an infection that is hidden, inactive, or dormant. As opposed to active infections, where a virus or bacterium is actively replicating and potentially causing symptoms, latent infections are essentially static.

Horizontal transmission

Horizontal transmission the spread of an infectious agent from one individual to another, usually through contact with bodily excretions or fluids, such as sputum or blood, that contain the agent.
vertical transmission transmission from one generation to another.

Vertical transmission

Vertical transmission: Passage of a disease-causing agent (pathogen) from mother to baby during the period immediately before and after birth. Transmission might occur across the placenta, in the breast milk, or through direct contact during or after birth. For example, HIV can be a vertically transmitted pathogen.

